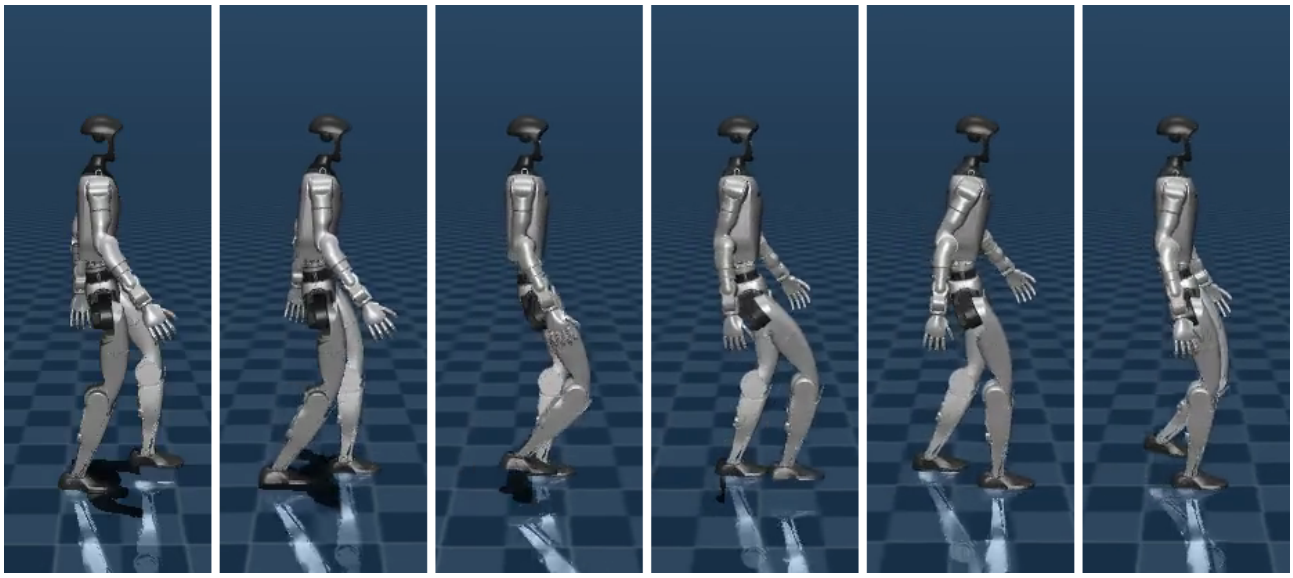
PROJECT SUMMARY

Teaching a Unitree G1 humanoid to walk with PPO in MuJoCo

A human-like gait benchmark (v1), a symmetric PPO baseline, and a DISC comparison

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Frozen v1 baseline (PPO seed 1, 16M steps) walking at 0.5 m/s, side view, six frames across one gait cycle.

At a glance

- Goal: velocity-tracking walking for the 29-DOF Unitree G1 humanoid in standard CPU MuJoCo, trained on a laptop (M2 Mac), with a gait that looks human-like, not just one that moves forward.
- Built a Gymnasium environment with a gait clock and ~48 reward terms; froze it as **benchmark v1** so RL algorithms can be compared fairly.
- Baseline: **SymmetricPPO** (PPO + left/right mirror loss). Two independent seeds both learn to walk from ~7–8M steps, reach the same gait at 16M steps (~1.5 h), with **0 falls in 18 benchmark episodes**, 97–99% phase match and speed within 0.01 m/s.
- Implemented **DISC** (Han & Sung, ICML 2019) on top of it. After tuning, DISC walks **1–2M steps earlier** than PPO on both seeds, but its final gait is less polished.

1. Goal and setup

The project trains a simulated Unitree G1 humanoid to walk forward at a commanded speed (0 to 1.2 m/s), including standing still when the command is 0. Beyond moving at the right speed, the target is a gait a person would recognise as natural: legs alternating on a steady rhythm, bent knees, flat-footed landings, feet placed under the hips, straight and symmetric leg paths, and arms swinging opposite the legs. The finished environment is frozen as a **benchmark** so that different RL algorithms can be compared on the same task.

Component	Choice
Robot model	Unitree G1 from MuJoCo Menagerie (29 joints, position actuators), loaded from `unitree_g1/scene.xml`
Simulator	Standard CPU MuJoCo 3.x (no GPU / MJX), physics step 0.005 s, policy at 50 Hz
RL library	Stable-Baselines3 PPO (subclassed), Gymnasium API, VecNormalize for observations and rewards
Hardware	Apple M2 MacBook Air; ~3,000 env steps/s with 8 parallel envs, so 16M steps ≈ 1.5 h
Starting point	A starter baseline (kept in `backup_original/`): fixed 0.5 m/s command, basic penalties, no foot-contact terms, explicitly not verified to walk

Code layout

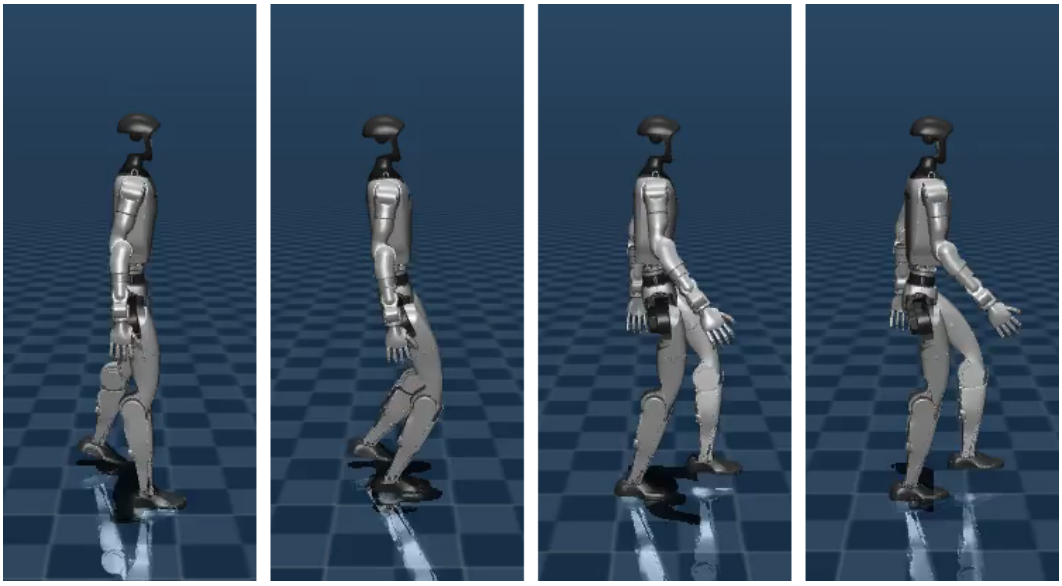
File	Lines	Role
g1_env.py	598	The environment: control, observations, gait clock, all reward terms, mirror maps
train.py	135	Training entry point: SymmetricPPO or DISC, VecNormalize, LR decay, checkpoints, resume, per-term reward logging
symmetric_ppo.py	134	PPO plus a left/right mirror symmetry loss
disc_ppo.py	228	DISC: dimension-wise importance-sampling clipping with off-policy replay
benchmark_eval.py	167	Standard evaluation: gait, foot, placement, arm and standing metrics over several policies
compare_runs.py	143	Learning curves and 'first walking checkpoint' detection across runs
render_video.py / evaluate.py	94	Offscreen MP4 rendering / interactive viewer

2. How the project evolved

The project went through three phases. Most of the learning came from watching videos of each policy, finding what looked wrong, and changing one or two things at a time.

Phase	What was done	Outcome
1. Curriculum approach (tag `old-approach`)	Separate stand → balance → gait → walk envs with a custom PyTorch PPO and policy transfer between stages; ~10k lines across many env, train and render scripts.	Hard to maintain and debug; replaced by a single environment.
2. Single env + SB3 PPO (23 Oct)	One velocity-tracking env, SB3 PPO, VecNormalize, checkpoints/resume, video rendering. Added the gait clock, phase-contact and swing-height rewards, heading hold, gait-scaled velocity reward.	First alternating gait at 4M steps, but with locked, hyperextended knees and feet lifted by hip hiking.
Bent-knee default	Replaced Menagerie's straight-legged keyframe with a bent-knee default pose (hip −0.3, knee +0.6, ankle −0.3 rad).	4M steps from scratch: 92% phase match, ~9 cm foot lift, knee 16–60°, no falls at 0–0.8 m/s.
Iterative refinement (23–24 Oct)	A chain of resumed runs adding one feature at a time: foot placement and flat feet, waist joints, stride length, torso twist, start-up (command ramp, double-support start), arm control and swing, then the mirror symmetry loss.	Each fixed a visible flaw, but the long resume chain left old habits in the policy.

Phase	What was done	Outcome
3. Benchmark v1 (24 Oct)	Froze the env as v1, added SymmetricPPO and benchmark_eval.py, retrained from scratch on two seeds.	Both seeds reproduce the same high-quality gait. Committed as the baseline (tag `benchmark-v1`).
DISC comparison (25–26 Oct)	Implemented DISC, verified it against SB3, tuned it on seed 1, confirmed on seed 2.	Tuned DISC learns to walk 1–2M steps sooner; final gait slightly worse.



Left to right: straight-leg policy (4M steps, locked knees); first bent-knee policy (4M); frozen v1 PPO baseline (16M); tuned DISC (16M). All at 0.5 m/s, side view.

3. The environment (benchmark v1)

The environment is a single Gymnasium class, `G1Env`. Any change to observations, actions, rewards, dynamics or episode setup must bump `ENV_VERSION`, and results across versions are never compared.

Aspect	Design
Control	50 Hz policy. Actions are PD position targets: default pose + action × per-joint range.
Joints	23 controlled: 12 leg joints, waist yaw/roll/pitch, shoulder pitch/roll/yaw and elbow per arm. Wrists hold their pose.
Default pose	Bent knees (pelvis at 0.76 m), because the straight-legged Menagerie keyframe let policies lock their knees. Arms hang with soft actuators (kp 50 instead of 500) so they swing like pendulums.
Commands	Forward speed sampled per episode from 0–1.2 m/s; 30% of episodes are standing (0 m/s). The command ramps in over 0.5 s; episodes start in double support with either leg leading.
Gait clock	Period 1.1 s at ≤ 0.3 m/s down to 0.8 s at ≥ 1.0 m/s. Leg signals left = sin(2π·phase), right = −sin(2π·phase); a leg is in stance while its signal is above −0.15 (short double support). Zero when standing.
Observations (96)	Body-frame linear and angular velocity, gravity direction, command, heading error, sin/cos of each leg's phase, all 29 joint positions and velocities, previous action.
Episode	1000 steps (20 s); a fall ends the episode with −200.

Reward design

Rewards follow Isaac Lab / legged_gym conceptually, but the weights were tuned for this setup. They fall into seven groups:

Group	Terms	Why it matters
Task	Velocity tracking, yaw rate, pelvis and torso heading, alive bonus, fall penalty	Velocity reward is **scaled by how well the feet follow the clock**, so shuffling earns only ~30% of it. Heading terms stop the policy walking in circles (velocity is measured in the body frame).
Gait	Feet on the clock, swing height ($\sin^2$ profile, 9 cm peak), swing reach, step length, air time, flight, left/right symmetry	Produces a clean alternating gait with steps that scale with speed (step length = command × period / 2).
Feet	Flat foot in stance and late swing, slip, landing speed, bounces, landing width (7 cm from centreline), no crossovers, step width	Removes 'catwalk' feet, toe-stubbing and stomping.
Posture	Knee bend (~17° in stance), hip yaw/roll, base height, torso upright, pelvis roll, waist, body acceleration, tilt	Avoids locked knees and lifting the foot by hiking or twisting the hip.
Arms	Clock-driven sinusoidal swing centred on vertical (±15° at 0.5 m/s to ±25° at 1.2 m/s), opposite the same-side leg; symmetric; elbow coupled to swing	Natural counter-swing instead of arms held stiff or flailing.
Standing	Feet down and still, no drift (weight 2.0)	Stands quietly at 0 m/s instead of marching in place.
Smoothness	Action rate and jerk, joint effort, arm joint speed	Smoother, lower-energy motion.

4. Training: Symmetric PPO

PPO kept settling into lopsided gaits (one leg doing more work, asymmetric swing paths) even with symmetry reward terms. The fix was **SymmetricPPO**, following Yu et al. (2018): an extra loss term asks that the policy's mean action on a state equals the mirrored mean action on the left/right-mirrored state. The mirror maps (which observation and action indices swap, and which flip sign) were verified exactly against a physically mirrored simulator state. Because observations are normalised, they are un-normalised, mirrored and re-normalised inside the loss.

Hyperparameter	Value	Hyperparameter	Value
Parallel envs × rollout	8 × 1024 steps	Network (actor & critic)	ELU MLP [512, 256, 128]
Minibatch / epochs	2048 / 5	Initial action std	≈ 0.37
Learning rate	3e-4 → 3e-5, linear decay	Symmetry loss weight	1.0
Budget	16M steps (~1.5 h)	Normalisation	VecNormalize (obs + reward)

The learning-rate schedule is also rescaled correctly on `--resume` (SB3 otherwise keeps the saved rate and measures progress against the total step count). Each run records its arguments, env version and git commit in `run_info.json`, plus per-episode Monitor CSVs and per-reward-term TensorBoard curves.

5. Evaluation methodology

Reward curves turned out to be a poor guide to gait quality: circling, marching in place, locked knees, catwalk feet and asymmetric swing only showed up in video or detailed metrics. So `benchmark_eval.py` defines a standard evaluation:

- Each policy runs 3 episodes × 20 s at each command (0, 0.5 and 1.0 m/s; reset seeds 0–2) with deterministic actions. Steady-state metrics skip the first 2 s.
- **Gait:** falls, speed and its error, phase match with the clock, touchdowns per gait cycle (1.0 = one clean touchdown per step), heading error.
- **Feet:** stance slip, landing speed, foot roll, flat 4-point contact, landing distance from the centreline per foot, crossovers, step length vs target, closest swing-foot approach to the centreline.

- **Body and arms:** pelvis roll, forward-speed surge around touchdown, arm swing back/forward, peak arm speed, elbow left/right difference.
- **Standing:** both-feet-down share, foot lift and drift at 0 m/s.
- Results are reported as mean/spread over several seeds with the same step budget, never from `ep_rew_mean` alone.

A 'walking standard' used to compare algorithms: no falls at 0 / 0.5 / 1.0 m/s, and at 0.5 and 1.0 m/s a speed error ≤ 0.1 m/s, phase match ≥ 95% and heading error ≤ 15°, holding at the next checkpoint as well.

6. Benchmark v1 results (PPO baseline)

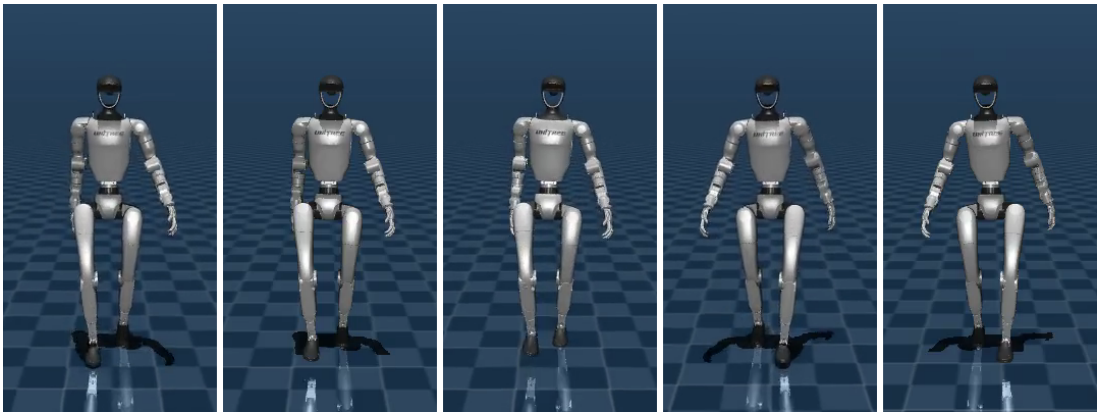
SymmetricPPO with default settings, trained from scratch for 16M steps on two seeds. Both walk with full 20 s episodes from about 7–8M steps and stay stable to the end. **0 falls in 18 benchmark episodes.**

Metric	0.5 m/s: seed 1 / seed 2	1.0 m/s: seed 1 / seed 2
Speed	0.489 / 0.496 m/s	0.998 / 1.005 m/s
Phase match	98.8% / 99.0%	97.8% / 97.0%
Touchdowns per gait cycle	1.02 / 1.02	1.14 / 1.07
Stance slip (median)	0.044 / 0.027 m/s	0.068 / 0.058 m/s
Landing speed (median)	0.20 / 0.14 m/s	0.37 / 0.16 m/s
Stance foot roll	1.3° / 0.7°	1.5° / 1.2°
Flat 4-point contact	68% / 71%	62% / 39%
Landing out from centreline (L / R)	6.3 / 6.5 cm, 6.3 / 6.6 cm	6.7 / 6.6 cm, 6.8 / 6.9 cm
Crossovers	0% / 0%	0% / 0%
Step length (target 0.254 / 0.400 m)	0.257 / 0.249 m	0.395 / 0.392 m
Max heading error	3.2° / 4.0°	4.1° / 8.5°
Pelvis roll range	4.5° / 4.4°	7.2° / 3.9°
Speed surge at touchdown	12% / 15%	11% / 8%
Arm swing back / forward	16° / 17°, 17° / 16°	26° / 25°, 27° / 26°
Elbow left/right difference	0.3° / 0.3°	0.8° / 0.8°

Standing (0 m/s), both seeds: both feet down 100% of the time, foot lift 0.16 cm, drift ≤ 2 mm.

What the numbers say

- Speed tracking is essentially exact (errors of 0.002–0.011 m/s) and the legs follow the clock 97–99% of the time.
- Foot placement is consistent and symmetric: ~6.5 cm out from the centreline on both sides, no crossovers, step length within 2% of target.
- Arms swing ±16–17° at 0.5 m/s and ±25–27° at 1.0 m/s, matching the targets, with left and right elbows within 1°.
- Two independent seeds give nearly identical numbers, so the result is reproducible rather than a lucky run.



Front view at 0.5 m/s: seed 1 (first three frames) and seed 2 (last two). Feet land under the hips with no crossovers; arms swing opposite the legs.

7. DISC vs PPO: sample efficiency

With a frozen benchmark in place, the next step was to test a different algorithm. **DISC** (Han & Sung, ICML 2019, *Dimension-Wise Importance Sampling Weight Clipping for Sample-Efficient RL*) changes three things relative to PPO:

- **Per-dimension clipping:** each action dimension's importance-sampling ratio is clipped separately, instead of clipping the product over all 23 dimensions. With high-dimensional actions, PPO's joint ratio leaves many samples with zero gradient.
- **Adaptive IS loss:** a penalty with coefficient α_{IS} keeps the update near a target divergence J_{targ}.
- **Off-policy replay:** recent rollouts are reused, with GAE-V advantages, when their per-dimension IS deviation is below ε_b.

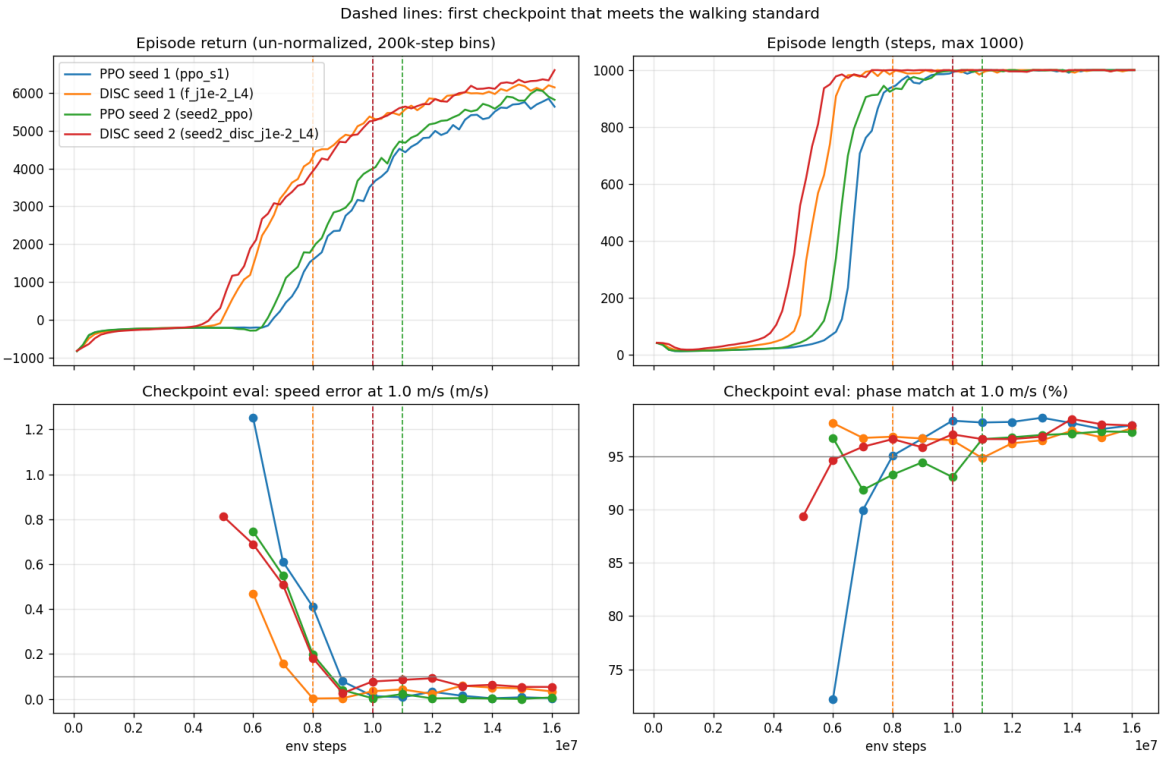
The implementation follows the authors' code and is built on SymmetricPPO, so the mirror loss applies as in the baseline. As a correctness check, on a single on-policy batch its log-probs, values, advantages and targets match SB3's GAE to float precision.

Tuning

With the paper's settings (ε 0.4, J_{targ} 1e-4, ε_b 0.1, 16 replayed rollouts) the IS loss held each update to ~1% of PPO's (approx. KL 0.0002–0.0005 vs ~0.02–0.045), α_{IS} sat at its cap, no dimension was ever clipped, and DISC **had not learned to walk after 16M steps**. A screen of 8M-step runs on seed 1 (only DISC tuned; PPO kept the benchmark settings):

DISC setting	Effect
J_{targ} 1e-3 → 1e-2	Largest single gain; without replay it tracks PPO's curve until PPO starts walking
32 gradient steps per epoch (paper) instead of 4	No gain, slower early
Replay 4 rollouts, J_{targ} 1e-2	**Best:** full-length episodes ~2M steps before PPO
J_{targ} 1e-1, no replay	Fast at first, then unstable over 16M steps (KL ~0.6); its deterministic policy falls
J_{targ} 3e-2, replay 4	Unstable (KL ~0.3)

Result (tuned DISC vs PPO, 16M steps, two seeds; seed 2 was not used for tuning)



Top: episode return and length vs env steps. Bottom: checkpoint evaluation at 1.0 m/s. Dashed lines mark the first checkpoint meeting the walking standard (orange/red = DISC, blue/green = PPO).

	PPO seed 1	DISC seed 1	PPO seed 2	DISC seed 2
First walking checkpoint	10M	**8M**	11M	**10M**
Episode return at 6M / 8M / 10M	−204 / 1306 / 3270	1095 / 4055 / 5254	−259 / 1689 / 3895	1577 / 3682 / 5110
Episode return at 16M	5811	6134	5977	6339
Samples with zero policy gradient	13–24%	~0%	13–24%	~0%
Wall-clock, 16M steps (2 runs in parallel)	3.6 h	4.9 h	3.4 h	4.7 h

Final gait quality at 16M steps: all four policies have 0 falls and equal standing, phase match and foot placement, but DISC's gait is less polished:

Metric	PPO, seeds 1 / 2	DISC, seeds 1 / 2
Speed error at 1.0 m/s	0.003 / 0.005 m/s	0.032 / 0.053 m/s
Flat contact at 0.5 m/s	71% / 72%	55% / 55%
Touchdowns per gait cycle at 1.0 m/s	1.10 / 1.07	1.25 / 1.32
Pelvis roll range at 0.5 m/s	4.1° / 4.4°	9.7° / 8.4°

Takeaways from the comparison

- Tuned DISC reaches each return level ~2M steps before PPO and meets the walking standard 1–2M steps sooner, on both seeds.
- The paper's hyperparameters did not transfer: the default J_{targ} made updates ~100× too small for this task.
- DISC uses its samples better but is ~35% slower in wall-clock time, and its action std collapses to ~0.05 (PPO ~0.11): it stops exploring earlier, which likely explains the rougher final gait. A higher entropy coefficient is the next thing to try.
- Caveat: two seeds is a small sample, and runs are not bit-for-bit reproducible (two seed-1 runs of one setting differed ~2× in return at 8M).

8. Lessons learned

Lesson	Detail
Look at the robot, not the reward curve	Circling, marching in place, locked knees, catwalk feet and asymmetric swing were all invisible in reward curves. Every change was judged with benchmark_eval.py plus front and side videos.
The default pose shapes the gait	With Menagerie's straight-legged keyframe, PPO locked and hyperextended its knees. A bent-knee default fixed it immediately.
Symmetry needs a loss, not just rewards	Symmetry reward terms still left lopsided gaits; the mirror symmetry loss in SymmetricPPO solved it.
Gate the task reward on gait quality	Scaling velocity reward by clock-following removed the shortcut of shuffling forward.
Body-frame velocity allows circling	A policy can satisfy body-frame velocity while walking in circles; heading must be rewarded and checked separately.
Long resume chains act as a hidden curriculum	Iterating by resuming under changing rewards left old habits in the policy; final quality came from from-scratch runs.
Keep evaluation windows inside one episode	The VecEnv auto-resets at step 1000; a longer rollout once recorded the reset as a phantom 'arm snap'.
Freeze and version the benchmark	Any env change bumps ENV_VERSION, and results are never compared across versions, so the PPO/DISC comparison is fair.
Published hyperparameters may not transfer	DISC with the paper's settings did not walk at all; one parameter (J_{targ}) made the difference.

9. Limitations and next steps

- A forward-speed surge of ~10–15% remains around each touchdown.
- Flat-foot contact at 1.0 m/s varies between seeds (39–62%).
- Flat ground only, with no pushes, terrain or randomised dynamics, so robustness to disturbances and sim-to-real transfer are untested.
- Forward walking only: no lateral or turning commands yet.
- DISC: try a higher entropy coefficient to keep exploring longer, and run more seeds for a stronger statistical comparison.
- Natural extensions: domain randomisation and push recovery (a v2 benchmark), turning commands, and other algorithms (e.g. SAC, off-policy PPO variants) on the frozen v1 env.

Appendix: reproducing the results

```
# M = --model ../mujoco_menagerie/unitree_g1/scene.xml
python train.py $M --steps 16000000 --envs 8 --seed 1 --out runs_seed1
python train.py $M --steps 16000000 --envs 8 --seed 1 --algo disc --j-targ 1e-2 --replay 4 --out
runs_disc_seed1
python benchmark_eval.py $M --policies checkpoints/g1_ppo_seed1_16M.zip
checkpoints/g1_ppo_seed2_16M.zip
python compare_runs.py $M --runs runs_seed1 runs_disc_seed1 --out compare.png
python render_video.py $M --policy checkpoints/g1_ppo_seed1_16M.zip --command 0.5 --azimuth 180
```

Baseline checkpoints: `checkpoints/g1_ppo_seed{1,2}_16M.zip`, each with its VecNormalize stats file (`*_vecnormalize.pkl`), which must sit next to the policy.